

EduConnect - Business Requirements Document (BRD)

Project Name: EduConnect - Career Guidance & College Admission Platform **Version:** 2.0 **Date:** 03 April 2026 **Original Concept Date:** 02 March 2026 **Team:** Abijith S (U19VU23S0026), Muhammad Ansar A H (U19VU23S0002), Abanas Krishna (U19VU23S0009) **Program:** BCA 6th Semester **Document Status:** Final Draft

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1. Executive Summary

EduConnect is a comprehensive, responsive web platform that simplifies the college admission process for students in India and Abroad. It provides centralized college listings, step-by-step admission guidance, career resources, and a personalized student dashboard — all accessible from desktop and mobile devices.

What the original presentation covered:

- Basic introduction, problem statement, 4 proposed solutions
- 5 system modules (College Listings, Admission Guidance, Career Guidance, Student Dashboard, Blog/Resources)
- Software & hardware requirements
- Tech stack: Django, HTML, MySQL

What was missing and has been added in this BRD:

- Detailed user personas and role-based access
- Authentication & authorization system
- Search, filter, and compare functionality
- Notification & alert system
- Scholarship & financial aid module
- Entrance exam information module
- College reviews & ratings system
- Counselor booking / live chat
- Admin panel & content management

- Responsive design specifications (mobile + desktop)
- Complete database schema design
- API endpoint specifications
- Security, SEO, and performance requirements
- Third-party integrations
- Analytics & reporting dashboard
- Detailed user flows
- Page-by-page wireframe breakdown
- Success metrics and KPIs
- Risk assessment and mitigation plan
- Project timeline with milestones

2. Problem Statement

Core Problems (from original presentation)

- Students face confusion during college admissions
- Lack of centralized information about colleges
- Difficulty in tracking applications
- Limited career guidance resources

Extended Problems Identified

#	Problem	Impact
P1	No single platform compares colleges across India and abroad side-by-side	Students visit 10+ websites to gather basic info
P2	Admission deadlines are scattered and easily missed	Students miss application windows
P3	Scholarship/financial aid information is fragmented	Students miss financial opportunities
P4	No personalized recommendations based on student profile	Students apply to wrong-fit colleges
P5	Entrance exam schedules and syllabus are hard to track	Exam preparation suffers
P6	Parents have no visibility into their child's application progress	Family decision-making is hampered
P7	Career guidance is generic, not tailored to individual strengths	Students make uninformed career choices
P8	Mobile access to admission tools is poor in existing platforms	70%+ students browse on mobile

3. Project Objectives

#	Objective	Measurable Target
O1	Centralize college information	500+ colleges listed within 6 months
O2	Simplify application tracking	Students can track all applications in one dashboard
O3	Provide personalized career guidance	AI-based recommendations based on student profile
O4	Ensure mobile-first responsive design	100% feature parity across mobile and desktop
O5	Deliver deadline alerts	Zero missed deadlines for active users
O6	Enable college comparison	Side-by-side comparison of up to 4 colleges
O7	Reduce admission confusion	80% user satisfaction rating

4. Target Audience & User Personas

4.1 Primary Users

Persona 1: Student (Age 16-24)

- **Who:** High school/undergraduate students seeking admission
- **Goals:** Find the right college, track applications, get career guidance
- **Pain Points:** Information overload, missed deadlines, no personalized advice
- **Device:** Primarily mobile (70%), desktop (30%)

Persona 2: Parent/Guardian

- **Who:** Parents helping children with college decisions
- **Goals:** Monitor child's application progress, compare college costs, understand ROI
- **Pain Points:** No visibility into the process, financial planning difficulty
- **Device:** Desktop (60%), mobile (40%)

4.2 Secondary Users

Persona 3: Career Counselor

- **Who:** Professional guidance counselors
- **Goals:** Manage multiple students, provide recommendations, track progress
- **Pain Points:** Manual tracking, no centralized tool
- **Device:** Desktop (80%)

Persona 4: College/Institution Representative

- **Who:** Admissions staff from listed colleges
- **Goals:** Update college info, respond to queries, manage listings
- **Pain Points:** Keeping information current across platforms
- **Device:** Desktop (90%)

4.3 System Users

Persona 5: Platform Admin

- **Who:** EduConnect team members
- **Goals:** Manage content, moderate reviews, monitor analytics, handle support
- **Device:** Desktop only

User Roles & Permissions Matrix

Feature	Student	Parent	Counselor	College Rep	Admin
View college listings	Yes	Yes	Yes	Yes	Yes
Create/Edit profile	Yes	Yes	Yes	Yes	Yes
Track applications	Yes	View only	View all assigned	No	View all
Save/Bookmark colleges	Yes	Yes	Yes	No	No
Write reviews	Yes	No	No	No	Moderate
Compare colleges	Yes	Yes	Yes	No	Yes
Book counselor session	Yes	Yes	N/A	No	Manage
Edit college info	No	No	No	Own college	All
Access admin panel	No	No	No	No	Yes

Receive notifications	Yes	Yes	Yes	Yes	Yes
View analytics	Own	Own	Assigned students	Own college	All

5. Scope

5.1 In Scope (MVP - Phase 1)

- User registration & authentication (email + Google OAuth)
- College listings with search, filter, and sort
- College detail pages with courses, fees, eligibility, deadlines
- Admission guidance content (step-by-step guides)
- Career guidance content and resources
- Student dashboard with profile, saved colleges, application tracker
- Blog/resource section
- Responsive design (mobile + desktop)
- Basic notification system (email)
- Contact/inquiry forms
- Admin panel for content management

5.2 In Scope (Phase 2)

- College comparison tool
- Scholarship & financial aid database
- Entrance exam module
- Reviews & ratings system
- Counselor booking system
- Parent dashboard (linked accounts)
- Push notifications
- Advanced search with AI recommendations

5.3 In Scope (Phase 3)

- Live chat / chatbot for instant guidance
- Video counseling integration
- College representative portal
- Analytics dashboard
- Community forum
- Mobile app (React Native / Flutter)

5.4 Out of Scope

- Direct application submission to colleges
- Payment processing for college fees
- Official document verification
- Visa processing assistance
- Accommodation booking

6. Functional Requirements

FR-01: User Authentication & Account Management

ID	Requirement	Priority
FR-01.1	User registration with email, name, phone, and password	Must Have
FR-01.2	Google OAuth / Social login	Should Have

FR-01.3	Email verification on registration	Must Have
FR-01.4	Password reset via email link	Must Have
FR-01.5	Role-based registration (Student, Parent, Counselor)	Must Have
FR-01.6	Profile creation wizard on first login	Must Have
FR-01.7	Profile edit and update	Must Have
FR-01.8	Account deactivation/deletion	Must Have
FR-01.9	Session management and auto-logout after inactivity	Should Have
FR-01.10	Parent-Student account linking via invite code	Should Have

FR-02: College Listings

ID	Requirement	Priority
FR-02.1	Display colleges in card/list view with key info	Must Have
FR-02.2	Search by college name, location, course	Must Have
FR-02.3	Filter by country, state, city, course type, fee range, ranking	Must Have
FR-02.4	Sort by ranking, fees, name, popularity	Must Have
FR-02.5	College detail page with: name, description, logo, images, location, contact	Must Have
FR-02.6	Courses offered with eligibility criteria per course	Must Have
FR-02.7	Fee structure breakdown (tuition, hostel, other)	Must Have
FR-02.8	Admission deadlines with countdown timer	Must Have
FR-02.9	Placement statistics (average package, top recruiters)	Should Have
FR-02.10	Campus facilities list	Should Have
FR-02.11	College location on embedded map	Should Have
FR-02.12	Photo gallery / virtual tour link	Could Have
FR-02.13	Save/bookmark college to dashboard	Must Have
FR-02.14	Share college page via link/social media	Should Have
FR-02.15	Pagination and infinite scroll on listing page	Must Have

FR-03: College Comparison

ID	Requirement	Priority
FR-03.1	Add up to 4 colleges to comparison tray	Should Have
FR-03.2	Side-by-side comparison table (fees, courses, ranking, placement, facilities)	Should Have
FR-03.3	Highlight differences and better values	Could Have
FR-03.4	Save/export comparison as PDF	Could Have

FR-04: Admission Guidance

ID	Requirement	Priority
FR-04.1	Step-by-step admission process guides per country/college type	Must Have
FR-04.2	Essay/SOP writing tips with sample templates	Must Have
FR-04.3	Interview preparation guides with common questions	Must Have
FR-04.4	Document checklist (customizable per college type)	Must Have
FR-04.5	Application timeline planner	Should Have
FR-04.6	LOR (Letter of Recommendation) guidance	Should Have
FR-04.7	Standardized test guides (SAT, GRE, GMAT, IELTS, TOEFL)	Should Have
FR-04.8	Country-specific admission guides (USA, UK, Canada, Australia, Germany)	Must Have

FR-05: Career Guidance

ID	Requirement	Priority
FR-05.1	Career path explorer (visual career maps)	Must Have
FR-05.2	Skill assessment questionnaire	Should Have
FR-05.3	Personalized career recommendations based on profile	Should Have
FR-05.4	Industry trends articles and data	Must Have
FR-05.5	Job market insights per field/location	Must Have
FR-05.6	Skill development resource links (courses, certifications)	Must Have
FR-05.7	Salary insights by role and experience	Should Have
FR-05.8	"Day in the life" professional profiles	Could Have
FR-05.9	Career quiz to match interests to professions	Should Have

FR-06: Student Dashboard

ID	Requirement	Priority
FR-06.1	Personalized student profile (academic details, interests, test scores)	Must Have
FR-06.2	Application tracker (college name, status, deadlines, documents)	Must Have
FR-06.3	Saved/bookmarked colleges list	Must Have
FR-06.4	Deadline calendar view with color coding	Must Have
FR-06.5	Notification center (all alerts in one place)	Must Have
FR-06.6	Progress indicator (profile completion, applications sent)	Should Have
FR-06.7	Recent activity feed	Should Have
FR-06.8	Quick actions (add application, browse colleges, get career advice)	Should Have
FR-06.9	Document upload and management	Should Have

FR-06.10	Notes/journal for each college application	Could Have
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FR-07: Scholarship & Financial Aid (NEW MODULE)

ID	Requirement	Priority
FR-07.1	Scholarship database with search and filter	Should Have
FR-07.2	Eligibility checker based on student profile	Should Have
FR-07.3	Scholarship deadlines and reminders	Should Have
FR-07.4	Education loan information from partner banks	Could Have
FR-07.5	Fee comparison calculator	Could Have

FR-08: Entrance Exams Module (NEW MODULE)

ID	Requirement	Priority
FR-08.1	Entrance exam database (JEE, NEET, CAT, SAT, GRE, etc.)	Should Have
FR-08.2	Exam dates, registration deadlines, syllabus	Should Have
FR-08.3	Preparation resources and tips	Should Have
FR-08.4	Previous year question papers / practice links	Could Have
FR-08.5	Exam-to-college mapping (which exams for which colleges)	Should Have

FR-09: Blog/Resource Section

ID	Requirement	Priority
FR-09.1	Blog articles categorized by topic	Must Have
FR-09.2	Search and filter blog posts	Must Have
FR-09.3	Featured/trending articles section	Should Have
FR-09.4	Author profiles for blog writers	Could Have
FR-09.5	Related articles suggestions	Should Have
FR-09.6	Social sharing for articles	Should Have
FR-09.7	Newsletter subscription	Should Have

FR-10: Notifications & Alerts

ID	Requirement	Priority
FR-10.1	Email notifications for deadline reminders	Must Have
FR-10.2	In-app notification center	Must Have
FR-10.3	Push notifications (browser)	Should Have
FR-10.4	Customizable notification preferences	Should Have
FR-10.5	Weekly digest email with relevant updates	Could Have

FR-11: Reviews & Ratings (NEW MODULE)

ID	Requirement	Priority
FR-11.1	Students can rate colleges (1-5 stars) on multiple parameters	Should Have
FR-11.2	Written reviews with moderation	Should Have
FR-11.3	Verified student badge for reviews	Could Have
FR-11.4	Helpful/unhelpful voting on reviews	Could Have
FR-11.5	Average rating display on college cards	Should Have

FR-12: Counselor Booking (NEW MODULE)

ID	Requirement	Priority
FR-12.1	Counselor profile listing with specialization	Should Have
FR-12.2	Booking calendar with available slots	Should Have
FR-12.3	Video call integration (Google Meet / Zoom link)	Could Have
FR-12.4	Session notes and follow-up tracking	Could Have
FR-12.5	Rating and feedback for counselors	Could Have

FR-13: Admin Panel

ID	Requirement	Priority
FR-13.1	Dashboard with platform statistics (users, colleges, traffic)	Must Have
FR-13.2	CRUD operations for colleges, courses, articles	Must Have
FR-13.3	User management (view, suspend, delete)	Must Have
FR-13.4	Content moderation (reviews, comments)	Must Have
FR-13.5	Bulk import colleges via CSV/Excel	Should Have
FR-13.6	SEO metadata management per page	Should Have
FR-13.7	Analytics and reporting dashboard	Should Have
FR-13.8	System logs and audit trail	Should Have

FR-14: Search (Global)

ID	Requirement	Priority
FR-14.1	Global search bar accessible from all pages	Must Have
FR-14.2	Search across colleges, courses, articles, exams	Must Have
FR-14.3	Auto-suggest / autocomplete	Should Have
FR-14.4	Search results grouped by category	Should Have
FR-14.5	Recent searches saved per user	Could Have

7. Non-Functional Requirements

NFR-01: Performance

ID	Requirement	Target
NFR-01.1	Page load time (First Contentful Paint)	< 1.5 seconds
NFR-01.2	Time to Interactive	< 3 seconds
NFR-01.3	API response time	< 500ms for 95th percentile
NFR-01.4	Support concurrent users	1,000+ simultaneous
NFR-01.5	Database query time	< 200ms average
NFR-01.6	Image optimization	WebP format, lazy loading
NFR-01.7	Lighthouse performance score	> 85

NFR-02: Responsiveness

ID	Requirement	Target
NFR-02.1	Mobile breakpoint	320px - 767px
NFR-02.2	Tablet breakpoint	768px - 1023px
NFR-02.3	Desktop breakpoint	1024px+
NFR-02.4	Touch-friendly tap targets	Minimum 44x44px
NFR-02.5	No horizontal scroll on any device	100% compliance
NFR-02.6	Tested browsers	Chrome, Safari, Firefox, Edge
NFR-02.7	Tested mobile OS	iOS 15+, Android 10+

NFR-03: Accessibility

ID	Requirement	Target
NFR-03.1	WCAG compliance level	AA minimum
NFR-03.2	Keyboard navigation	Full support
NFR-03.3	Screen reader compatibility	ARIA labels on all interactive elements
NFR-03.4	Color contrast ratio	4.5:1 minimum
NFR-03.5	Alt text on all images	100% compliance

NFR-04: Scalability

ID	Requirement	Target
NFR-04.1	Horizontal scaling support	Stateless backend
NFR-04.2	CDN for static assets	Mandatory
NFR-04.3	Database connection pooling	Configured

NFR-04.4	Caching strategy	Redis/Memcached for frequent queries
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NFR-05: Availability

ID	Requirement	Target
NFR-05.1	Uptime	99.5%
NFR-05.2	Planned maintenance window	< 2 hours/month
NFR-05.3	Backup frequency	Daily automated backups
NFR-05.4	Recovery Time Objective (RTO)	< 4 hours

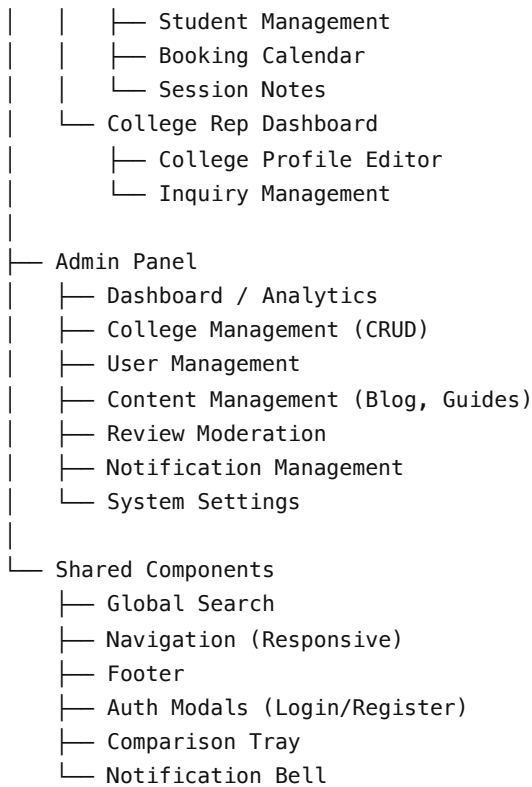
NFR-06: Internationalization

ID	Requirement	Target
NFR-06.1	Default language	English
NFR-06.2	Future language support ready	i18n framework in place
NFR-06.3	Currency display	INR and USD
NFR-06.4	Date format	DD/MM/YYYY (user configurable)
NFR-06.5	Timezone handling	User timezone auto-detected

8. System Modules (Detailed)

Module Architecture Overview

EduConnect Platform	
└─ Public Pages (No auth required)	
└─ Home / Landing Page	
└─ College Listings (Browse)	
└─ College Detail Page	
└─ Admission Guides	
└─ Career Guidance Hub	
└─ Blog / Resources	
└─ Entrance Exams Info	
└─ Scholarship Database	
└─ About Us	
└─ Contact Us	
└─ Authenticated Pages	
└─ Student Dashboard	
└─ Profile Management	
└─ Application Tracker	
└─ Saved Colleges	
└─ Deadline Calendar	
└─ Notification Center	
└─ Document Manager	
└─ Parent Dashboard	
└─ Linked Student View	
└─ Progress Monitor	
└─ Counselor Dashboard	



Module Details

M1: College Listings Module

Purpose: Centralized, searchable database of colleges in India and abroad.

Data per College:

- Basic: Name, Logo, Type (Government/Private/Deemed), Established Year
- Location: Country, State, City, Address, Map Coordinates
- Academics: Courses offered, Departments, Faculty count
- Admissions: Eligibility criteria per course, Deadlines, Required exams, Fee structure
- Placements: Average package, Highest package, Top recruiters, Placement %
- Facilities: Hostel, Library, Labs, Sports, WiFi, Canteen
- Rankings: NIRF, QS, THE, country-specific rankings
- Media: Photos, Videos, Virtual tour link
- Contact: Phone, Email, Website, Social media links
- Reviews: Average rating, Number of reviews

M2: Admission Guidance Module

Purpose: Step-by-step guides for application processes.

Content Structure:

- Country-wise guides (India, USA, UK, Canada, Australia, Germany, etc.)
- Degree-level guides (Undergraduate, Postgraduate, PhD, Diploma)
- Application process walkthrough
- Essay/SOP writing tips with templates
- Interview preparation (common questions, video tips)
- Document checklist generator
- LOR request templates
- Test preparation guides (SAT, GRE, GMAT, IELTS, TOEFL, JEE, NEET, CAT)
- Timeline planner tool

M3: Career Guidance Module

Purpose: Help students make informed career decisions.

Content Structure:

- Career path explorer (interactive visual career maps)
- Career quiz / aptitude assessment
- Industry-wise career information
- Required skills per career path
- Salary ranges by role, experience, and location
- Skill development resources (online courses, certifications)
- Industry trends and reports
- Job market insights
- Professional day-in-the-life profiles

M4: Student Dashboard Module

Purpose: Personalized control center for each student.

Features:

- Profile: Academic history, test scores, interests, preferred locations
- Application Tracker: Kanban-style board (Researching → Applying → Applied → Accepted → Enrolled)
- Saved Colleges: Bookmarked colleges with quick compare
- Deadline Calendar: Monthly/weekly view with color-coded deadlines
- Notifications: In-app alerts for deadlines, new colleges, blog updates
- Documents: Upload, organize, and manage application documents
- Progress: Profile completion %, application progress

M5: Blog/Resource Section

Purpose: Content hub for educational articles and resources.

Features:

- Categories: Admission tips, Career advice, Study abroad, Exam prep, Student life
- Tags for cross-referencing
- Featured/trending articles
- Author profiles
- Related article suggestions
- Social sharing
- Newsletter signup

M6: Scholarship & Financial Aid (NEW)

Purpose: Aggregate scholarship opportunities and financial planning tools.

Features:

- Searchable scholarship database
- Filter by: Country, Degree, Field, Amount, Eligibility
- Auto-match scholarships to student profile
- Deadline tracking with reminders
- Education loan information
- Cost of living calculator by city

M7: Entrance Exams (NEW)

Purpose: Central hub for exam information and preparation.

Features:

- Exam database: JEE, NEET, CAT, SAT, GRE, GMAT, IELTS, TOEFL, etc.
- Exam dates and registration deadlines
- Syllabus and exam pattern
- Preparation tips and resources
- Exam-to-college mapping
- Previous year papers (links)

M8: Reviews & Ratings (NEW)

Purpose: Peer-driven college feedback system.

Features:

- Multi-parameter ratings (Academics, Placement, Infrastructure, Faculty, Campus Life)
- Written reviews with text
- Moderation queue for admin
- Verified student badge
- Helpful/unhelpful voting

M9: Counselor Booking (NEW)

Purpose: Connect students with professional guidance counselors.

Features:

- Counselor profiles (specialization, experience, ratings)
- Calendar-based slot booking
- Video call link generation
- Session notes
- Follow-up scheduling

9. User Flows

UF-01: New Student Registration & Onboarding

Landing Page → Click "Sign Up"

→ Registration Form (Name, Email, Phone, Password, Role: Student)

→ Email Verification

→ Profile Setup Wizard:

- Step 1: Academic Details (Current education level, Marks/GPA, Board/University)
- Step 2: Interests (Preferred fields, Career interests)
- Step 3: Preferences (Preferred countries, Budget range, Degree level)
- Step 4: Test Scores (Optional: SAT, GRE, IELTS, etc.)

→ Dashboard (with recommended colleges based on profile)

UF-02: College Discovery & Comparison

Home/Navbar → Click "Colleges" or use Search Bar

→ College Listing Page (card view with filters sidebar)

→ Apply Filters (Country: India, Course: B.Tech, Fee: Under 5L)

→ Browse Results → Click College Card

→ College Detail Page (tabs: Overview, Courses, Fees, Admissions, Placements, Reviews)

→ Click "Add to Compare" (max 4) OR "Save to Dashboard"

→ Click "Compare" button → Comparison Page (side-by-side table)

→ Click "Apply Guidance" → Admission Guide for that college type

UF-03: Application Tracking

Dashboard → Click "Add New Application"
→ Select College (from saved or search)
→ Set Status (Researching)
→ Add Deadline Date
→ Upload Required Documents
→ Move through stages: Researching → Preparing → Applied → Under Review → Accepted/Rejected
→ System sends reminders 30, 7, 3, 1 days before deadline

UF-04: Career Exploration

Navbar → Click "Career Guidance"
→ Career Hub Page → Click "Take Career Quiz"
→ Answer 15-20 questions about interests, skills, values
→ Get Results: Top 5 Career Matches with compatibility %
→ Click a Career → Career Detail Page (description, skills needed, salary, growth, related courses)
→ Click "Find Colleges for this Career" → Filtered college listing

UF-05: Blog & Resource Consumption

Navbar → Click "Resources/Blog"
→ Blog Listing Page (filter by category/tag)
→ Click Article → Full Article Page
→ Related Articles shown at bottom
→ Click "Share" → Social media share options
→ Click "Subscribe" → Newsletter signup modal

UF-06: Counselor Booking

Dashboard → Click "Book a Counselor"
→ Counselor Listing (filter by specialization, rating)
→ Click Counselor Profile → View details, reviews
→ Click "Book Session" → Calendar with available slots
→ Select Date & Time → Confirm Booking
→ Receive Email with Video Call Link
→ Post-session: Rate counselor

10. Page-by-Page Breakdown (Responsive Web)

P01: Landing Page / Home

Purpose: First impression, explain value, drive sign-ups

Desktop Layout:

- Hero section: Full-width banner with tagline, search bar, CTA buttons ("Explore Colleges" / "Sign Up Free")
- Stats bar: X+ Colleges, Y+ Students, Z+ Countries
- Featured Colleges carousel (6 cards visible)
- How it Works section (3-step visual guide)
- Popular Categories grid (Engineering, Medical, MBA, Arts, Study Abroad)
- Testimonials slider
- Latest Blog Posts (3 cards)
- Newsletter signup banner
- Footer (links, social media, contact)

Mobile Layout:

- Hamburger menu
- Hero section: Stacked layout, prominent search bar, single CTA
- Stats bar: 2x2 grid
- Featured Colleges: Horizontal scrollable cards
- How it Works: Vertical stacked steps
- Categories: 2-column grid
- Single testimonial with swipe
- 1 blog card with "View All" link
- Sticky bottom nav: Home, Search, Dashboard, Menu

P02: College Listing Page

Desktop Layout:

- Top: Breadcrumb, search bar with autocomplete
- Left sidebar: Filters (country, state, course, fee range sliders, ranking, college type checkboxes)
- Right: Results count, sort dropdown, view toggle (grid/list)
- College cards: Image, name, location, rating, top courses, fee range, CTA buttons (View / Compare / Save)
- Pagination at bottom

Mobile Layout:

- Search bar at top
- Filter button → Opens bottom sheet with filters
- Sort button → Dropdown
- Cards: Stacked full-width, compact layout
- Infinite scroll instead of pagination
- Floating "Compare (X)" button when colleges added

P03: College Detail Page

Desktop Layout:

- Hero: College banner image, logo, name, location, rating, quick stats
- Sticky tab navigation: Overview | Courses | Fees | Admissions | Placements | Reviews | Gallery
- Overview tab: Description, key facts grid, location map, contact info
- Courses tab: Accordion list with eligibility per course
- Fees tab: Table with breakdown
- Admissions tab: Deadlines, required exams, process steps
- Placements tab: Stats, top recruiters logos, salary chart
- Reviews tab: Average rating breakdown, review cards, write review CTA
- Sidebar: Quick actions (Save, Compare, Apply Guidance, Share), Similar Colleges

Mobile Layout:

- Scrollable tabs (horizontal)
- Collapsed sidebar actions → Sticky bottom action bar (Save, Compare, Share)
- Accordion sections for dense content
- Full-width tables with horizontal scroll

P04: College Comparison Page

Desktop Layout:

- 2-4 column layout with college headers
- Rows: Rating, Fees, Courses, Ranking, Placement %, Avg Package, Facilities, Deadlines
- Highlighted best value per row
- Remove/swap college buttons
- Print/PDF export button

Mobile Layout:

- 2-column max, swipe for more
- Sticky college name headers
- Collapsible category rows

P05: Admission Guidance Hub

Desktop Layout:

- Hero: "Your Admission Journey Starts Here"
- Category cards: By Country, By Degree Level, By Exam
- Featured guides carousel
- Step-by-step process timeline (visual)
- Download checklist CTA

Mobile Layout:

- Vertical card stack
- Expandable timeline steps
- Sticky "Download Checklist" button

P06: Career Guidance Hub

Desktop Layout:

- Hero with "Take Career Quiz" CTA
- Career path explorer (interactive tree/map)
- Industry cards grid
- Trending careers section
- Salary insights charts
- Skill development resources

Mobile Layout:

- Simplified career path (list view instead of tree)
- Swipeable industry cards
- Collapsible sections

P07: Student Dashboard

Desktop Layout:

- Left sidebar: Navigation (Profile, Applications, Saved, Calendar, Notifications, Documents)
- Main content area switches per section
- Top bar: Welcome message, profile completion %, notification bell
- Application Tracker: Kanban board with drag-and-drop
- Calendar: Monthly view with deadline dots
- Quick Stats cards at top

Mobile Layout:

- Bottom tab navigation (replaces sidebar)
- Application Tracker: List view with status badges (Kanban too complex for mobile)
- Calendar: Agenda/list view by default
- Pull-to-refresh on all sections

P08: Blog/Resources Page

Desktop Layout:

- Featured article hero (large card)
- Category tabs/pills
- 3-column article grid

- Sidebar: Popular posts, categories, tags cloud, newsletter signup

Mobile Layout:

- Single column card list
- Category horizontal scroll pills
- No sidebar (integrated into main flow)

P09: Scholarship Page

Desktop Layout:

- Search + filter (similar to college listing)
- Scholarship cards: Name, amount, eligibility summary, deadline, apply link
- Eligibility checker tool in sidebar

Mobile Layout:

- Bottom sheet filters
- Stacked cards
- Eligibility checker as separate tab

P10: Entrance Exams Page

Desktop Layout:

- Exam category tabs (Engineering, Medical, Management, Abroad)
- Exam cards: Name, date, registration deadline, quick links
- Upcoming exams timeline
- Preparation resources section

Mobile Layout:

- Swipeable category tabs
- Compact exam cards
- Countdown badges for upcoming exams

P11: Login / Register Pages

Desktop Layout:

- Split screen: Left = branding/illustration, Right = form
- Social login buttons at top
- Form fields with inline validation
- Terms & conditions checkbox

Mobile Layout:

- Full-width form
- Social login buttons prominent
- Minimal branding (logo only)

P12: Contact / About Pages

Desktop Layout:

- About: Team section, mission, vision, stats
- Contact: Form + map + contact details side-by-side

Mobile Layout:

- Stacked sections
- Click-to-call phone numbers
- Click-to-email addresses

P13: Admin Panel

Desktop Only (minimum 1024px):

- Left sidebar navigation
- Dashboard with charts (users, traffic, colleges, popular searches)
- Data tables with search, filter, sort, bulk actions
- CRUD modals for content management
- Review moderation queue

11. Tech Stack

Frontend (Updated from original)

Component	Technology	Reason
Framework	React.js 18+ with Next.js 14+	SEO-friendly SSR, fast routing, responsive by design
Styling	Tailwind CSS 3+	Utility-first, responsive classes built-in, smaller bundle
UI Components	shadcn/ui	Accessible, customizable, professional components
State Management	React Context + TanStack Query	Server state caching, minimal client state
Forms	React Hook Form + Zod	Performant forms with schema validation
Charts	Recharts	Dashboard analytics visualization
Calendar	React Big Calendar	Deadline calendar in dashboard
Rich Text Editor	TipTap	Blog/guide content editing in admin
Icons	Lucide React	Consistent, lightweight icon set

Backend (Aligned with original)

Component	Technology	Reason
Framework	Python Django 5+	Original team expertise, robust ORM, admin panel
API	Django REST Framework	RESTful API for React frontend
Authentication	django-allauth + JWT	Social auth + token-based auth for SPA
Task Queue	Celery + Redis	Email notifications, deadline reminders
Search	django-filter + PostgreSQL Full Text	Advanced filtering and search
File Storage	Django Storages + AWS S3 / Cloudinary	Document & image uploads
CORS	django-cors-headers	Cross-origin for separate frontend

Database (Updated from original)

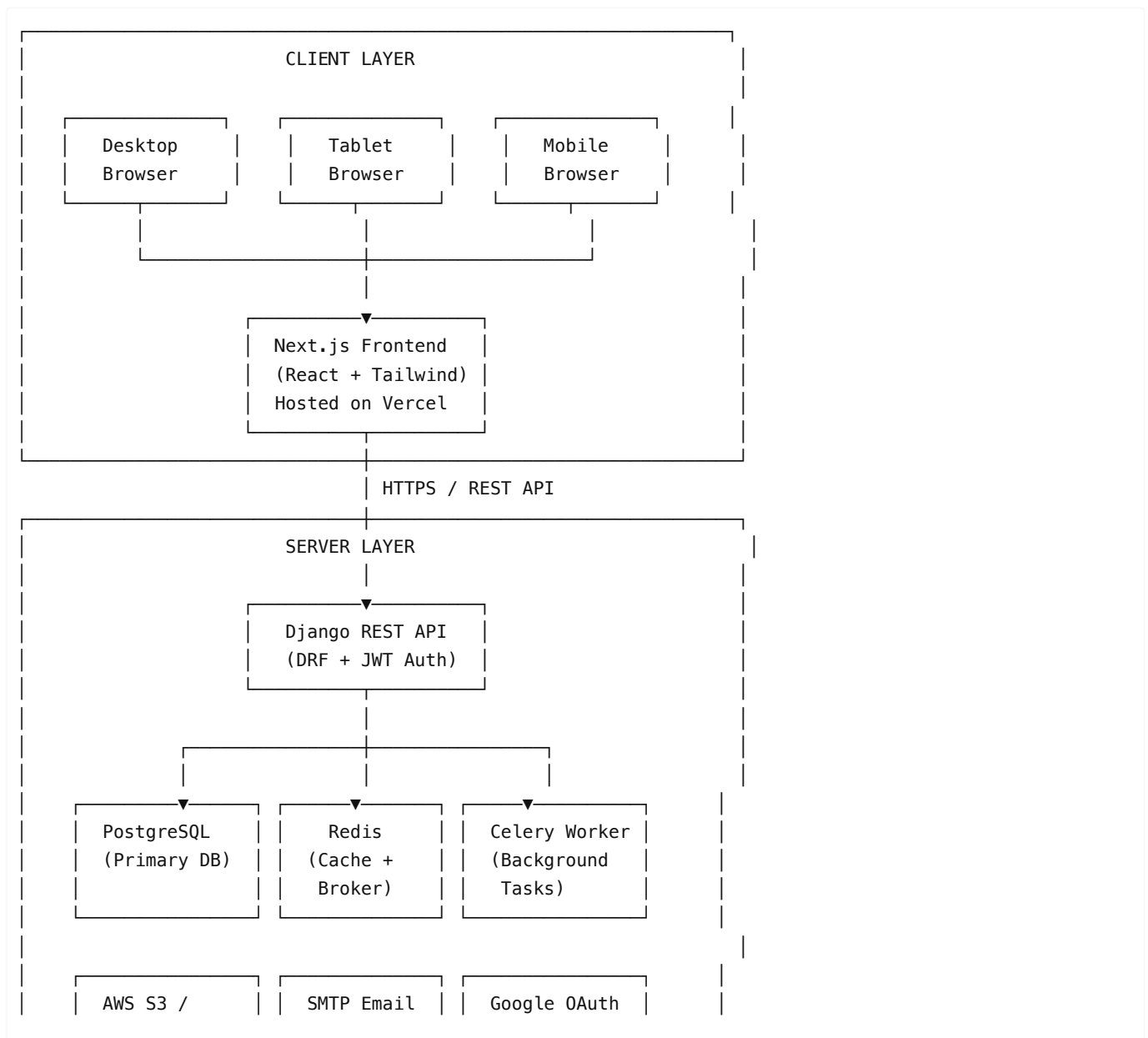
Component	Technology	Reason
Primary DB	PostgreSQL 15+	Superior to MySQL for JSON, full-text search, array fields
Cache	Redis	Session cache, query cache, Celery broker
Search Index	PostgreSQL GIN indexes	Full-text search without extra infra (Phase 1)

Note: Original spec used MySQL. PostgreSQL is recommended for production due to better JSON support, full-text search, and array fields. MySQL can be used if team prefers, with minimal changes.

DevOps & Infrastructure

Component	Technology	Reason
Hosting (Frontend)	Vercel	Zero-config Next.js deployment, global CDN
Hosting (Backend)	Railway / Render / AWS EC2	Django deployment with managed PostgreSQL
Version Control	Git + GitHub	Collaboration and CI/CD
CI/CD	GitHub Actions	Automated testing and deployment
Monitoring	Sentry	Error tracking
Analytics	Google Analytics 4 + PostHog	User behavior analytics

12. System Architecture



Cloudinary (File Store)	Service (SendGrid)	Provider
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13. Database Schema Design

Core Tables

users

```
users
├─ id (PK, UUID)
├─ email (UNIQUE, NOT NULL)
├─ password_hash (NOT NULL)
├─ first_name (VARCHAR 100)
├─ last_name (VARCHAR 100)
├─ phone (VARCHAR 15)
├─ role (ENUM: student, parent, counselor, college_rep, admin)
├─ avatar_url (VARCHAR 500)
├─ is_email_verified (BOOLEAN, DEFAULT false)
├─ is_active (BOOLEAN, DEFAULT true)
├─ created_at (TIMESTAMP)
└─ updated_at (TIMESTAMP)
```

student_profiles

```
student_profiles
├─ id (PK, UUID)
├─ user_id (FK → users.id, UNIQUE)
├─ date_of_birth (DATE)
├─ gender (ENUM: male, female, other, prefer_not_to_say)
├─ current_education_level (ENUM: high_school, undergraduate, graduate)
├─ institution_name (VARCHAR 200)
├─ board_university (VARCHAR 200)
├─ gpa_percentage (DECIMAL 5,2)
├─ preferred_countries (JSON ARRAY)
├─ preferred_fields (JSON ARRAY)
├─ preferred_degree_level (ENUM: ug, pg, phd, diploma)
├─ budget_min (DECIMAL 12,2)
├─ budget_max (DECIMAL 12,2)
├─ budget_currency (VARCHAR 3, DEFAULT 'INR')
├─ test_scores (JSON: {sat: 1400, gre: 320, ielts: 7.5, ...})
├─ bio (TEXT)
├─ profile_completion_pct (INT, DEFAULT 0)
├─ created_at (TIMESTAMP)
└─ updated_at (TIMESTAMP)
```

colleges

```
colleges
├─ id (PK, UUID)
├─ name (VARCHAR 300, NOT NULL)
├─ slug (VARCHAR 300, UNIQUE)
└─ type (ENUM: government, private, deemed, autonomous)
```

- └─ established_year (INT)
- └─ description (TEXT)
- └─ logo_url (VARCHAR 500)
- └─ banner_url (VARCHAR 500)
- └─ website (VARCHAR 500)
- └─ email (VARCHAR 200)
- └─ phone (VARCHAR 20)
- └─ country (VARCHAR 100, NOT NULL)
- └─ state (VARCHAR 100)
- └─ city (VARCHAR 100)
- └─ address (TEXT)
- └─ latitude (DECIMAL 10,8)
- └─ longitude (DECIMAL 11,8)
- └─ ranking_nirf (INT)
- └─ ranking_qs (INT)
- └─ ranking_the (INT)
- └─ accreditation (VARCHAR 200)
- └─ avg_package_lpa (DECIMAL 8,2)
- └─ highest_package_lpa (DECIMAL 8,2)
- └─ placement_percentage (DECIMAL 5,2)
- └─ total_students (INT)
- └─ total_faculty (INT)
- └─ campus_area_acres (DECIMAL 8,2)
- └─ facilities (JSON ARRAY: ["hostel","library","wifi",...])
- └─ social_links (JSON: {facebook, twitter, instagram, linkedin, youtube})
- └─ is_featured (BOOLEAN, DEFAULT false)
- └─ is_active (BOOLEAN, DEFAULT true)
- └─ avg_rating (DECIMAL 3,2, DEFAULT 0)
- └─ total_reviews (INT, DEFAULT 0)
- └─ created_at (TIMESTAMP)
- └─ updated_at (TIMESTAMP)

courses

courses

- └─ id (PK, UUID)
- └─ college_id (FK → colleges.id)
- └─ name (VARCHAR 200, NOT NULL)
- └─ degree_level (ENUM: ug, pg, phd, diploma, certificate)
- └─ department (VARCHAR 200)
- └─ duration_years (DECIMAL 3,1)
- └─ fee_per_year (DECIMAL 12,2)
- └─ fee_currency (VARCHAR 3, DEFAULT 'INR')
- └─ total_fee (DECIMAL 12,2)
- └─ eligibility_criteria (TEXT)
- └─ min_percentage (DECIMAL 5,2)
- └─ required_exams (JSON ARRAY)
- └─ seats_available (INT)
- └─ specializations (JSON ARRAY)
- └─ is_active (BOOLEAN, DEFAULT true)
- └─ created_at (TIMESTAMP)
- └─ updated_at (TIMESTAMP)

admission_deadlines

admission_deadlines

- └ id (PK, UUID)
- └ college_id (FK → colleges.id)
- └ course_id (FK → courses.id, NULLABLE)
- └ academic_year (VARCHAR 10: "2026-27")
- └ application_open_date (DATE)
- └ application_close_date (DATE)
- └ exam_date (DATE, NULLABLE)
- └ result_date (DATE, NULLABLE)
- └ counseling_start_date (DATE, NULLABLE)
- └ notes (TEXT)
- └ created_at (TIMESTAMP)
- └ updated_at (TIMESTAMP)

applications (Student Application Tracker)

applications

- └ id (PK, UUID)
- └ student_id (FK → users.id)
- └ college_id (FK → colleges.id)
- └ course_id (FK → courses.id, NULLABLE)
- └ status (ENUM: researching, preparing, applied, under_review, accepted, rejected, enrolled, withdrawn)
- └ deadline_date (DATE)
- └ applied_date (DATE, NULLABLE)
- └ result_date (DATE, NULLABLE)
- └ notes (TEXT)
- └ priority (ENUM: dream, target, safety)
- └ created_at (TIMESTAMP)
- └ updated_at (TIMESTAMP)

saved_colleges

saved_colleges

- └ id (PK, UUID)
- └ user_id (FK → users.id)
- └ college_id (FK → colleges.id)
- └ created_at (TIMESTAMP)
- └ UNIQUE (user_id, college_id)

reviews

reviews

- └ id (PK, UUID)
- └ user_id (FK → users.id)
- └ college_id (FK → colleges.id)
- └ rating_overall (INT 1-5)
- └ rating_academics (INT 1-5)
- └ rating_placements (INT 1-5)
- └ rating_infrastructure (INT 1-5)
- └ rating_faculty (INT 1-5)
- └ rating_campus_life (INT 1-5)
- └ review_text (TEXT)
- └ is_verified (BOOLEAN, DEFAULT false)
- └ is_approved (BOOLEAN, DEFAULT false)

```
|— helpful_count (INT, DEFAULT 0)
|— created_at (TIMESTAMP)
└— updated_at (TIMESTAMP)
```

blog_posts

```
blog_posts
|— id (PK, UUID)
|— author_id (FK → users.id)
|— title (VARCHAR 300)
|— slug (VARCHAR 300, UNIQUE)
|— excerpt (VARCHAR 500)
|— content (TEXT / Rich HTML)
|— cover_image_url (VARCHAR 500)
|— category (ENUM: admission_tips, career_advice, study_abroad, exam_prep, student_life, scholarships)
|— tags (JSON ARRAY)
|— is_featured (BOOLEAN, DEFAULT false)
|— is_published (BOOLEAN, DEFAULT false)
|— published_at (TIMESTAMP)
|— views_count (INT, DEFAULT 0)
|— created_at (TIMESTAMP)
└— updated_at (TIMESTAMP)
```

scholarships

```
scholarships
|— id (PK, UUID)
|— name (VARCHAR 300)
|— provider (VARCHAR 300)
|— description (TEXT)
|— amount (DECIMAL 12,2)
|— currency (VARCHAR 3)
|— eligibility_criteria (TEXT)
|— applicable_countries (JSON ARRAY)
|— applicable_degree_levels (JSON ARRAY)
|— applicable_fields (JSON ARRAY)
|— deadline (DATE)
|— application_url (VARCHAR 500)
|— is_active (BOOLEAN, DEFAULT true)
|— created_at (TIMESTAMP)
└— updated_at (TIMESTAMP)
```

entrance_exams

```
entrance_exams
|— id (PK, UUID)
|— name (VARCHAR 200)
|— full_name (VARCHAR 500)
|— category (ENUM: engineering, medical, management, law, abroad, government)
|— conducting_body (VARCHAR 300)
|— description (TEXT)
|— exam_date (DATE)
|— registration_open (DATE)
|— registration_close (DATE)
|— result_date (DATE)
```

```
|— syllabus_url (VARCHAR 500)
|— official_website (VARCHAR 500)
|— preparation_tips (TEXT)
|— is_active (BOOLEAN, DEFAULT true)
|— created_at (TIMESTAMP)
└— updated_at (TIMESTAMP)
```

notifications

```
notifications
|— id (PK, UUID)
|— user_id (FK → users.id)
|— type (ENUM: deadline_reminder, new_college, application_update, blog_post, scholarship, system)
|— title (VARCHAR 200)
|— message (TEXT)
|— link (VARCHAR 500)
|— is_read (BOOLEAN, DEFAULT false)
|— created_at (TIMESTAMP)
└— read_at (TIMESTAMP, NULLABLE)
```

counselors

```
counselors
|— id (PK, UUID)
|— user_id (FK → users.id, UNIQUE)
|— specialization (JSON ARRAY: ["study_abroad", "engineering", "mba"])
|— experience_years (INT)
|— bio (TEXT)
|— hourly_rate (DECIMAL 8,2)
|— avg_rating (DECIMAL 3,2, DEFAULT 0)
|— total_sessions (INT, DEFAULT 0)
|— is_available (BOOLEAN, DEFAULT true)
|— created_at (TIMESTAMP)
└— updated_at (TIMESTAMP)
```

counselor_bookings

```
counselor_bookings
|— id (PK, UUID)
|— counselor_id (FK → counselors.id)
|— student_id (FK → users.id)
|— session_date (DATE)
|— start_time (TIME)
|— end_time (TIME)
|— status (ENUM: scheduled, completed, cancelled, no_show)
|— meeting_link (VARCHAR 500)
|— notes (TEXT)
|— rating (INT 1-5, NULLABLE)
|— feedback (TEXT, NULLABLE)
|— created_at (TIMESTAMP)
└— updated_at (TIMESTAMP)
```

documents


```
documents
├─ id (PK, UUID)
├─ user_id (FK → users.id)
├─ application_id (FK → applications.id, NULLABLE)
├─ name (VARCHAR 200)
├─ file_url (VARCHAR 500)
├─ file_type (VARCHAR 50)
├─ file_size_bytes (BIGINT)
├─ category (ENUM: transcript, resume, sop, lor, certificate, id_proof, photo, other)
├─ created_at (TIMESTAMP)
└─ updated_at (TIMESTAMP)
```

parent_student_links

```
parent_student_links
├─ id (PK, UUID)
├─ parent_id (FK → users.id)
├─ student_id (FK → users.id)
├─ invite_code (VARCHAR 20, UNIQUE)
├─ status (ENUM: pending, active, revoked)
├─ created_at (TIMESTAMP)
└─ accepted_at (TIMESTAMP, NULLABLE)
```

14. API Endpoints

Authentication

Method	Endpoint	Description
POST	/api/auth/register/	Register new user
POST	/api/auth/login/	Login (returns JWT)
POST	/api/auth/logout/	Logout (blacklist token)
POST	/api/auth/refresh/	Refresh JWT token
POST	/api/auth/verify-email/	Verify email with token
POST	/api/auth/forgot-password/	Request password reset
POST	/api/auth/reset-password/	Reset password with token
GET	/api/auth/google/	Google OAuth redirect
GET	/api/auth/google/callback/	Google OAuth callback

Users & Profiles

Method	Endpoint	Description
GET	/api/users/me/	Get current user profile
PUT	/api/users/me/	Update current user profile
GET	/api/users/me/student-profile/	Get student profile
PUT	/api/users/me/student-profile/	Update student profile

DELETE	/api/users/me/	Deactivate account
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Colleges

Method	Endpoint	Description
GET	/api/colleges/	List colleges (with search, filter, sort, pagination)
GET	/api/colleges/:slug/	Get college detail
GET	/api/colleges/:slug/courses/	Get courses for a college
GET	/api/colleges/:slug/deadlines/	Get deadlines for a college
GET	/api/colleges/:slug/reviews/	Get reviews for a college
GET	/api/colleges/featured/	Get featured colleges
GET	/api/colleges/compare/?ids=1,2,3	Compare colleges

Saved Colleges

Method	Endpoint	Description
GET	/api/saved-colleges/	List saved colleges
POST	/api/saved-colleges/	Save a college
DELETE	/api/saved-colleges/:college_id/	Unsave a college

Applications

Method	Endpoint	Description
GET	/api/applications/	List user's applications
POST	/api/applications/	Create new application
PUT	/api/applications/:id/	Update application
PATCH	/api/applications/:id/status/	Update application status
DELETE	/api/applications/:id/	Delete application

Reviews

Method	Endpoint	Description
POST	/api/reviews/	Submit a review
PUT	/api/reviews/:id/	Edit own review
DELETE	/api/reviews/:id/	Delete own review
POST	/api/reviews/:id/helpful/	Mark review as helpful

Blog

Method	Endpoint	Description
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GET	/api/blog/	List published posts (filter by category, tag)
GET	/api/blog/:slug/	Get post detail
GET	/api/blog/featured/	Get featured posts
GET	/api/blog/categories/	List categories with post count

Scholarships

Method	Endpoint	Description
GET	/api/scholarships/	List scholarships (with filters)
GET	/api/scholarships/:id/	Get scholarship detail
GET	/api/scholarships/recommended/	Get recommended based on profile

Entrance Exams

Method	Endpoint	Description
GET	/api/exams/	List exams (with filters)
GET	/api/exams/:id/	Get exam detail
GET	/api/exams/upcoming/	Get upcoming exams

Notifications

Method	Endpoint	Description
GET	/api/notifications/	List user notifications
PATCH	/api/notifications/:id/read/	Mark as read
POST	/api/notifications/read-all/	Mark all as read
GET	/api/notifications/unread-count/	Get unread count

Counselors

Method	Endpoint	Description
GET	/api/counselors/	List counselors
GET	/api/counselors/:id/	Get counselor detail
GET	/api/counselors/:id/availability/	Get available slots
POST	/api/counselors/:id/book/	Book a session
POST	/api/counselor-bookings/:id/cancel/	Cancel booking
POST	/api/counselor-bookings/:id/rate/	Rate completed session

Documents

Method	Endpoint	Description
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GET	/api/documents/	List user documents
POST	/api/documents/	Upload document
DELETE	/api/documents/:id/	Delete document

Search

Method	Endpoint	Description
GET	/api/search/?q=term	Global search across all entities
GET	/api/search/autocomplete/?q=term	Autocomplete suggestions

Admin (Admin-only)

Method	Endpoint	Description
POST	/api/admin/colleges/	Create college
PUT	/api/admin/colleges/:id/	Update college
DELETE	/api/admin/colleges/:id/	Delete college
POST	/api/admin/colleges/bulk-import/	Bulk import from CSV
GET	/api/admin/users/	List all users
PATCH	/api/admin/users/:id/	Update user status
GET	/api/admin/reviews/pending/	Get pending reviews
PATCH	/api/admin/reviews/:id/approve/	Approve review
PATCH	/api/admin/reviews/:id/reject/	Reject review
GET	/api/admin/analytics/dashboard/	Get dashboard stats
POST	/api/admin/blog/	Create blog post
PUT	/api/admin/blog/:id/	Update blog post

15. Third-Party Integrations

Integration	Purpose	Priority
Google OAuth 2.0	Social login	Must Have
SendGrid / Resend	Transactional emails (verification, reminders)	Must Have
Cloudinary / AWS S3	Image & document storage	Must Have
Google Maps Embed API	College location maps	Should Have
Google Analytics 4	Website traffic analytics	Must Have
Google reCAPTCHA v3	Bot protection on forms	Should Have
Sentry	Error monitoring and alerting	Should Have
PostHog	Product analytics (user behavior)	Could Have

Google Meet / Zoom API	Counselor video sessions	Could Have (Phase 2)
Razorpay / Stripe	Counselor session payments (future)	Could Have (Phase 3)

16. Hardware & Software Requirements

Development Environment

Component	Requirement
OS	Windows 10/11, macOS 12+, Linux (Ubuntu 20.04+)
Processor	Intel i3 / AMD Ryzen 3 or higher
RAM	8 GB minimum (4 GB absolute minimum)
Storage	50 GB free space
Internet	10 Mbps or higher
IDE	VS Code (with ESLint, Prettier, Python, Tailwind plugins) or PyCharm
Runtime	Node.js 18+ (frontend), Python 3.11+ (backend)
Database	PostgreSQL 15+ (local)
Cache	Redis 7+ (local or Docker)
Browser	Chrome 100+, Firefox 100+, Safari 16+, Edge 100+
Version Control	Git 2.30+
Containerization	Docker (optional but recommended)

Production Server Requirements

Component	Minimum	Recommended
CPU	2 vCPU	4 vCPU
RAM	4 GB	8 GB
Storage	50 GB SSD	100 GB SSD
Bandwidth	100 GB/month	Unlimited
SSL Certificate	Required (Let's Encrypt free)	Required
OS	Ubuntu 22.04 LTS	Ubuntu 22.04 LTS

17. Security Requirements

#	Requirement	Implementation
S1	Authentication	JWT tokens with refresh rotation, httpOnly cookies
S2	Password Security	bcrypt hashing, minimum 8 chars, complexity rules
S3	Input Validation	Server-side validation on all inputs, Zod schemas on frontend

S4	SQL Injection Prevention	Django ORM (parameterized queries), no raw SQL
S5	XSS Prevention	React auto-escaping, Content Security Policy headers
S6	CSRF Protection	Django CSRF middleware, SameSite cookies
S7	Rate Limiting	django-ratelimit on auth and API endpoints
S8	HTTPS	Enforced via HSTS headers, redirect HTTP to HTTPS
S9	File Upload Validation	Type checking, size limits (5MB), malware scan
S10	CORS	Whitelist only frontend domain
S11	Data Encryption	TLS in transit, AES-256 at rest for sensitive data
S12	Secrets Management	Environment variables, never in code
S13	Dependency Scanning	GitHub Dependabot, pip-audit, npm audit
S14	Admin Access	2FA required for admin accounts
S15	Privacy	GDPR-compliant data handling, user data export/delete

18. SEO & Performance Requirements

SEO

#	Requirement
SEO-1	Server-side rendering (SSR) for all public pages
SEO-2	Unique meta title, description, and OG tags per page
SEO-3	Dynamic sitemap.xml generation
SEO-4	robots.txt configuration
SEO-5	Structured data (JSON-LD) for colleges, articles, reviews
SEO-6	Canonical URLs on all pages
SEO-7	Clean URL slugs (e.g., /colleges/iit-bombay)
SEO-8	Image alt tags on all images
SEO-9	Internal linking strategy
SEO-10	301 redirects for changed URLs

Performance

#	Requirement
P-1	Lighthouse score > 85 for Performance, Accessibility, SEO
P-2	Image optimization: WebP format, responsive sizes, lazy loading
P-3	Code splitting per route
P-4	Static page generation for stable content (blog posts, guides)

P-5	API response caching with Redis (TTL: 5 min for listings, 1 hr for static)
P-6	Database query optimization with indexes on searchable columns
P-7	Gzip/Brotli compression
P-8	CDN for static assets (Vercel Edge Network)
P-9	Bundle size budget: < 200KB initial JS
P-10	Core Web Vitals: LCP < 2.5s, FID < 100ms, CLS < 0.1

19. Success Metrics & KPIs

User Acquisition

Metric	Target (6 months)	Target (1 year)
Registered users	5,000	25,000
Monthly active users	2,000	10,000
Daily active users	500	3,000

Engagement

Metric	Target
Avg session duration	> 4 minutes
Pages per session	> 3
Return visitor rate	> 40%
Dashboard usage (weekly)	> 60% of registered users
College saves per user	> 5 average
Applications tracked per user	> 3 average

Content

Metric	Target (6 months)
Colleges listed	500+ (India) + 200+ (Abroad)
Blog posts published	100+
Admission guides	20+ country/degree combos
Scholarships listed	200+

Technical

Metric	Target
Page load time	< 2 seconds
API uptime	99.5%

Error rate	< 0.5%
Mobile traffic share	> 60%

Business

Metric	Target
User satisfaction (NPS)	> 50
Counselor booking rate	> 5% of active users
Newsletter subscribers	> 20% of registered users

20. Risk Assessment

#	Risk	Probability	Impact	Mitigation
R1	Inaccurate college data	High	High	Partner with colleges for verified data; admin review process; user-reported errors
R2	Low initial user adoption	Medium	High	SEO strategy, social media marketing, college partnerships, referral program
R3	Data scraping by competitors	Medium	Medium	Rate limiting, CAPTCHA, terms of service, legal protection
R4	Server downtime during peak admission season	Medium	High	Auto-scaling, load balancing, CDN, monitoring alerts
R5	Security breach / data leak	Low	Critical	Security audit, penetration testing, encrypted data, regular backups
R6	Scope creep delaying launch	High	Medium	Strict MVP definition, phased releases, weekly sprint reviews
R7	Team member unavailability	Medium	Medium	Code documentation, pair programming, knowledge sharing
R8	Third-party API changes/deprecation	Low	Medium	Abstraction layers, fallback mechanisms, monitoring
R9	Mobile performance issues	Medium	High	Mobile-first design, performance budgets, testing on real devices
R10	Content going stale (outdated deadlines)	High	High	Automated deadline alerts for admins, crowdsourced updates, annual audit cycle

21. Project Timeline & Milestones

Phase 1: MVP (Weeks 1-8)

Week	Milestone	Deliverables
1-2	Project Setup & Design	Repo setup, Figma designs, DB schema finalized, dev environment
3-4	Auth + College Listings	Registration, login, college CRUD, listing page with search/filter

5-6	Dashboard + Content	Student dashboard, application tracker, admission & career guides, blog
7	Integration + Polish	Notifications, saved colleges, responsive testing, bug fixes
8	Testing + Deploy	QA testing, performance optimization, MVP deployment

Phase 2: Enhanced Features (Weeks 9-14)

Week	Milestone	Deliverables
9-10	Comparison + Scholarships	College comparison, scholarship database, entrance exams
11-12	Reviews + Advanced Search	Reviews & ratings, AI recommendations, advanced filters
13-14	Parent + Counselor	Parent dashboard, counselor booking, push notifications

Phase 3: Scale & Optimize (Weeks 15-20)

Week	Milestone	Deliverables
15-16	Chat + Community	Live chat / chatbot, community forum
17-18	College Portal + Analytics	College rep dashboard, analytics dashboard
19-20	Optimization + Growth	Performance optimization, SEO, marketing integrations

22. Future Enhancements

#	Enhancement	Description
FE-1	Mobile App	React Native / Flutter app for iOS and Android
FE-2	AI Chatbot	GPT-powered admission assistant for instant answers
FE-3	Virtual Campus Tours	360-degree campus tours integrated on college pages
FE-4	Peer Mentorship	Connect current students with aspiring students
FE-5	Visa Guidance Module	Country-specific visa application guidance
FE-6	Accommodation Finder	Hostel/PG/apartment listings near colleges
FE-7	Education Loan Marketplace	Compare and apply for education loans
FE-8	Alumni Network	Connect with alumni for career insights
FE-9	Mock Interview AI	AI-powered mock interview practice
FE-10	Multi-language Support	Hindi, Tamil, Telugu, Kannada, Malayalam
FE-11	Offline Mode (PWA)	Progressive Web App for offline content access
FE-12	WhatsApp Notifications	Deadline reminders via WhatsApp Business API

Appendix A: Glossary

Term	Definition
BRD	Business Requirements Document

MVP	Minimum Viable Product
SSR	Server-Side Rendering
JWT	JSON Web Token
ORM	Object-Relational Mapping
CRUD	Create, Read, Update, Delete
API	Application Programming Interface
CDN	Content Delivery Network
WCAG	Web Content Accessibility Guidelines
SOP	Statement of Purpose
LOR	Letter of Recommendation
NPS	Net Promoter Score
LPA	Lakhs Per Annum

Appendix B: Document Revision History

Version	Date	Author	Changes
1.0	02 March 2026	Team EduConnect	Original presentation/abstract
2.0	03 April 2026	Team EduConnect	Full BRD with all missing requirements, responsive design specs, database schema, API design, security, SEO, timeline

This document serves as the complete specification for building EduConnect as a responsive web application. All development, design, and testing should reference this BRD as the single source of truth.